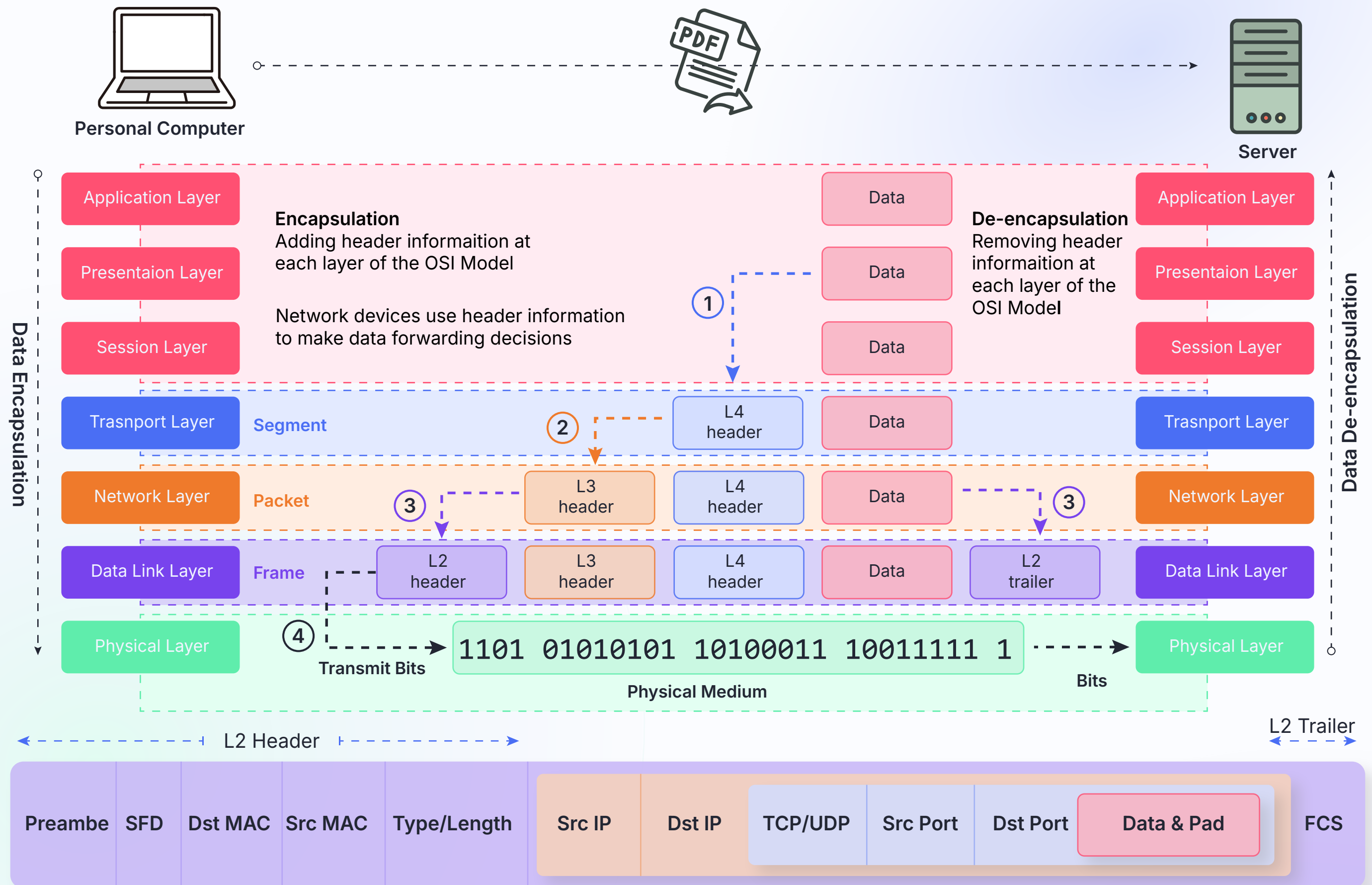


Data flow across OSI Model Layers



Ethernet Frame

- OSI Model Layer 2 Header
 - Data-Link Layer
 - Destination and Source Mac Address
 - Preamble- For device to sync their receiver clocks
 - SFD - Start Frame Delimiter, marks the end of a preamble and the start of the rest of the frame.
 - Type - Type of the encapsulated frame (IPv4 or IPv6)
 - Length - Length of the encapsulated packet
 - FCS - Frame Check Sequence, for error detection using CRC algorithm.
- Encapsulates all higher layer data into a frame

Packet

- OSI Model Layer 3 Header
 - Network Layer Protocol
 - Source IP Address
 - Destination IP Address
- Encapsulates all higher layer data into a packet

Segment

- OSI Model Layer 4 Header
- Transport Layer Protocol
 - TCP/UDP
 - Source Port
 - Destination Port
 - Encapsulates all higher layer data into a segment